



KidneyStrip Agent

Low-cost AI-guided urine ACR screening for early kidney-risk detection.

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1. Problem Statement & Why It Matters



Screening-support concept
Not a diagnostic or final
decision tool.

- ✓ Chronic kidney disease (CKD) is often a silent disease and many people are diagnosed late.
- ✓ Urine albumin-to-creatinine ratio (uACR) is an important early screening test for kidney damage, especially in people with diabetes and hypertension.
- ✓ Albuminuria increases risk of kidney failure and cardiovascular disease.
- ✓ Screening is underused and many patients are not tested early.



GLOBAL BURDEN

~**674 million**
people worldwide
have chronic kidney
disease.

WHO 2026 estimate



INDIA BURDEN

~**13–15%**
CKD prevalence
in India based on
studies.



DIABETES / HIGH-RISK POPULATION

~**89.8 million**
adults in India
have diabetes.

IDF 2024



SILENT DISEASE

CKD is often
asymptomatic until
advanced stages.
Many are diagnosed
too late.



THE PROBLEM

Current kidney-risk screening is
too late, too expensive, and too
dependent on clinic/lab infrastructure.

- ✗ People are tested only after symptoms or complications appear.
- ✗ uACR testing is underused in primary care and screening programs.
- ✗ Cost and lab dependence limit access, especially in low-resource settings.



THE IMPACT

Late detection leads to
dialysis, kidney failure,
higher treatment cost,
and preventable suffering.



More dialysis and
kidney failure



Higher long-term
treatment cost



Preventable suffering
and reduced quality of life



Key references summarized from WHO 2026, KDIGO 2024, IDF 2024, and India CKD studies.



2. Current Screening: Cost, Friction & Why It Fails

Current Lab-based uACR Pathway



Friction at Every Step



Cost of Current Screening

Current screening pathway (approx.):

₹600–₹1,500+ per screening



Lab analyzer equipment (approx.):

₹10–₹25 lakh class equipment

Often several lakhs for immunoturbidimetry analyzers



Turnaround Time

Current lab flow:

Hours to Days



Limitations of Current Approach

- ✗ Expensive
- ✗ Infrastructure-dependent
- ✗ Not portable
- ✗ Limited access in rural settings
- ✗ Repeat screening is poor
- ✗ Delayed detection



Only about 1 in 5 high-risk patients get routine albuminuria / uACR testing.

Based on a 2025 systematic review.

Why this problem happens



CKD is silent

No symptoms until advanced stages.



Screening is underused

Low awareness, low priority, and low repeat rates.



Current testing requires

Clinics, labs, equipment, and trained staff.



The current ask:
affordable, repeatable kidney-risk screening at the point of care and at home.





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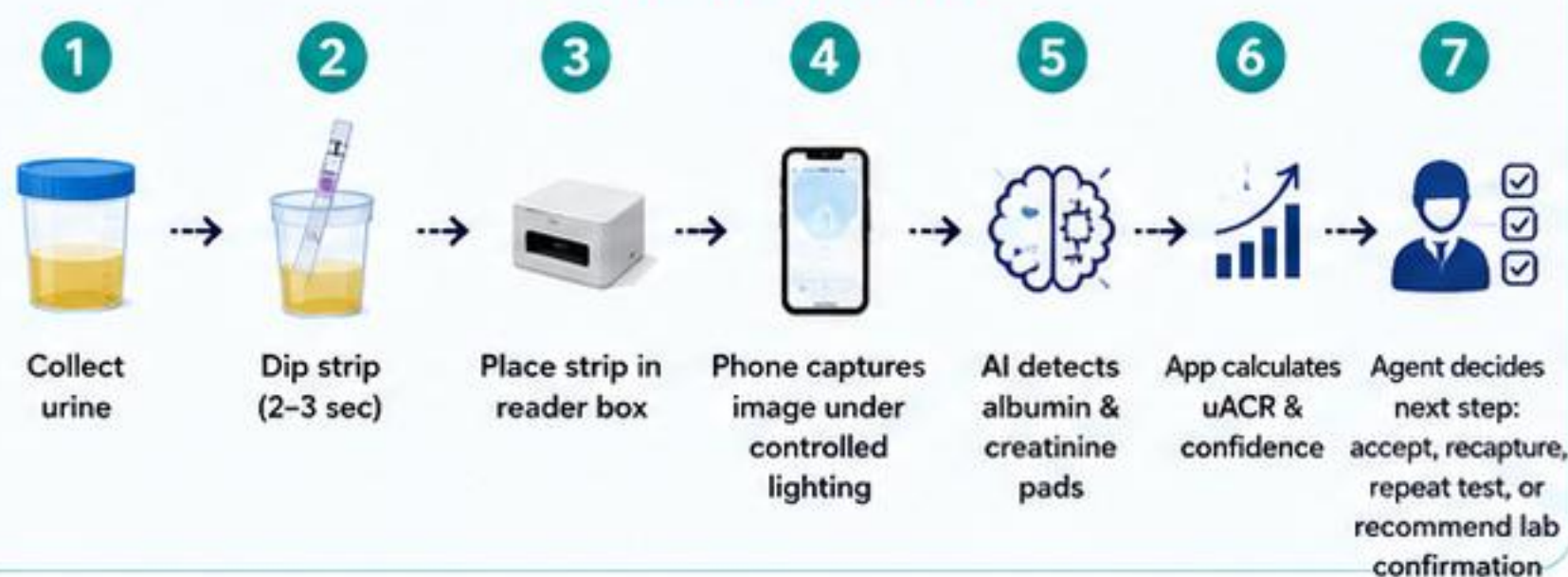
3. Our Solution — KidneyStrip Agent

KidneyStrip Agent combines an albumin + creatinine urine test strip, a small LED reader, and an AI app to deliver fast, accurate, and affordable uACR screening at the point of care.

HARDWARE / PHYSICAL SETUP



7-STEP WORKFLOW



COST ADVANTAGE



Our target test cost:
~₹20–₹40
per test

One-time low-cost reader hardware:
~₹1,000–₹3,000+

KEY BENEFITS



2-Minute Result



Works Offline



Portable & Lightweight



Easy to Use
(3 Simple Steps)



Suitable for
PHCs, Screening
Camps, Pharmacies
& Home Use

INNOVATION HIGHLIGHTS



- ✓ Controlled lighting ensures consistent, reliable readings.
- ✓ Image-quality check prevents false results from poor images.
- ✓ Uncertainty-aware reading with confidence score.
- ✓ Smart repeat-test recommendation for borderline results.
- ✓ Longitudinal tracking to monitor trends over time.

SAFETY NOTE



KidneyStrip Agent is a screening-support tool.
It is not a final diagnosis.
Abnormal results should be confirmed with laboratory tests and clinician evaluation.

BUILT FOR REAL-WORLD IMPACT



- ✓ Enables early detection of kidney risk.
- ✓ Supports timely intervention and better outcomes.
- ✓ Affordable, scalable, and sustainable.
- ✓ Designed for last-mile healthcare delivery.



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4. Real-World POC — What It Looks Like



4-STEP WORKFLOW



Portable & low-cost

- Small, light, and battery-friendly
- Est. device cost: < ₹2,000
- Per-test cost: < ₹30



Built for PHCs, screening camps, and community use

- Easy to use by health workers
- Works offline
- Scales for population screening

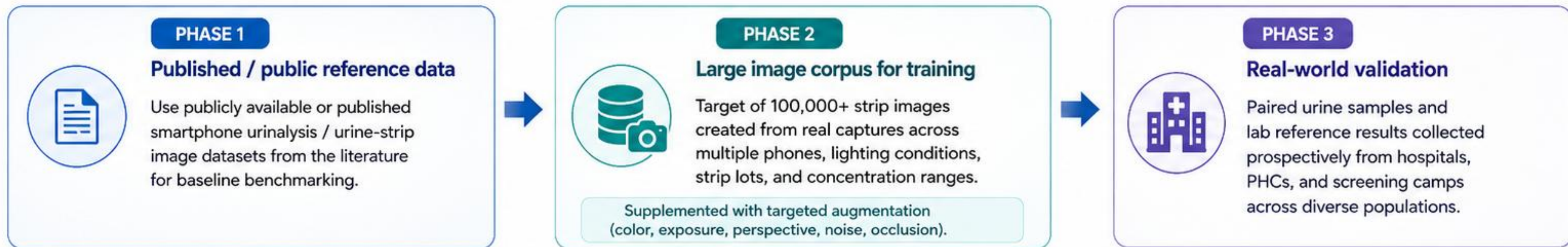


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5. How We Will Measure Accuracy

A rigorous, multi-phase benchmarking and validation plan to ensure clinical credibility and real-world reliability.



	PHASE 1: Published / public reference data	PHASE 2: Large image corpus for training	PHASE 3: Real-world validation
Data source	Published datasets from literature (e.g., smartphone urinalysis datasets, strip image datasets)	Our planned training corpus (100,000+ real-world strip images)	Prospective collection from hospitals, PHCs, and screening camps
Purpose	Baseline benchmarking and method comparison	Model training, calibration and generalization	Independent real-world validation and clinical performance assessment
Sample type	Published strip images (smartphone captures)	Urine-strip images from real captures across devices, lighting, lots, concentration ranges	Paired urine samples with urine-strip images
Ground truth	As provided in the published datasets (e.g., lab values or assigned labels)	From lab-tested subset used for calibration and quality control	Standard lab ACR / albumin and creatinine reference methods



ACCURACY METRICS

- ✓ Pad detection accuracy
- ✓ Concentration error / MAE
- ✓ ACR category accuracy (A1 / A2 / A3)
- ✓ Sensitivity & specificity for elevated ACR ≥ 30 mg/g
- ✓ Repeatability (intra- & inter-run)
- ✓ Cross-device robustness (phones, lighting, lots)



TESTING AGAINST REAL DATA

We will compare app output against lab reference values and report confusion matrix, ROC, and Bland-Altman / agreement style analysis.



Our system will be tested on holdout data never seen during training.

This ensures an unbiased estimate of real-world performance and generalizability.





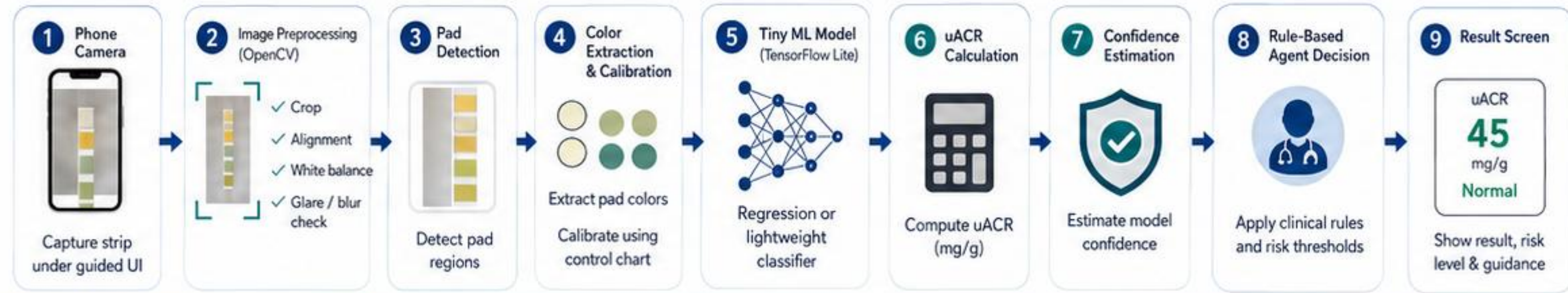
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6. Tech Stack & TinyML on Edge

KidneyStrip Agent runs the full AI pipeline on the smartphone. Images are processed locally. **No internet required.**



TECH STACK (ON-DEVICE)



Cross-platform app with guided UI & workflow



TinyML inference engine for on-device models



Image processing, preprocessing & computer vision



Local encrypted storage for results & history



Secure cloud sync, backup & analytics (opt-in)



Future hardware control for strip readers

MODEL SPEC & DEPLOYMENT



TinyML / lightweight models optimized for on-device inference.



Processing time under ~1 second after capture.



Uses smartphone CPU / NPU / GPU acceleration when available.

WHY EDGE AI?



Works offline
No internet required for screening.



Low latency
Instant results at point of care.



Privacy-first
All data stays on device by default.

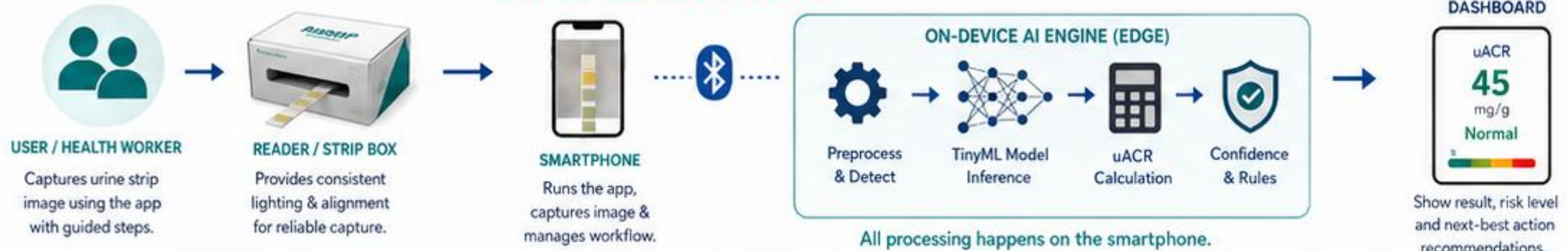


Low power
Optimized models for efficient inference.



No dependency on internet
Reliable in low-connectivity settings.

END-TO-END ARCHITECTURE



**Affordable.
Accurate.
Accessible.**

All processing can happen **on-device**.



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